21-242 Recitation 12

Dimension — Q7. (U+V)= U+ V- (U V)

Course-level solution, with a later-course alternative when useful

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Prompt

Show that $\dim(U+V) = \dim(U) + \dim(V) - \dim(U \cap V)$.

Course-level solution

Assume $U, V \subseteq W$ are finite-dimensional (e.g. if $\dim W < \infty$). Let $(\vec{z}_1, \ldots, \vec{z}_r)$ be a basis of $U \cap V$. Extend it to a basis $(\vec{z}_1, \ldots, \vec{z}_r, \vec{u}_1, \ldots, \vec{u}_s)$ of U and to a basis $(\vec{z}_1, \ldots, \vec{z}_r, \vec{v}_1, \ldots, \vec{v}_t)$ of V. We claim

$$\mathcal{B} = (\vec{z}_1, \ldots, \vec{z}_r, \vec{u}_1, \ldots, \vec{u}_s, \vec{v}_1, \ldots, \vec{v}_t)$$

is a basis of $U + V$.

Spanning. A typical vector of $U + V$ is $\vec{u} + \vec{v}$ with $\vec{u} \in U$, $\vec{v} \in V$, hence is a combination of the $\vec{z}$'s, $\vec{u}$'s, and $\vec{v}$'s.

Independence. If $\sum a_i \vec{z}_i + \sum b_j \vec{u}_j + \sum c_k \vec{v}_k = \vec{0}$, then $\sum c_k \vec{v}_k = -\sum a_i \vec{z}_i - \sum b_j \vec{u}_j \in U$, so $\sum c_k \vec{v}_k \in U \cap V = \mathrm{Span}(\vec{z}_i)$. But $(\vec{z}_1, \ldots, \vec{z}_r, \vec{v}_1, \ldots, \vec{v}_t)$ is independent, so every $c_k = 0$ and then every $a_i, b_j = 0$ by independence of the U-basis.

Thus $\dim(U + V) = r + s + t$, while $\dim U = r + s$, $\dim V = r + t$, $\dim(U \cap V) = r$, which is the formula.

Later-course solution (using ideas from the rest of the handout)

Map $U \times V \to U + V$ by $(\vec{u}, \vec{v}) \mapsto \vec{u} + \vec{v}$. It is surjective with kernel $\{(x, -x) : x \in U \cap V\} \cong U \cap V$. Rank-nullity on $U \times V$ (whose dimension is $\dim U + \dim V$) yields the same identity. This is the slick end-of-handout proof.